

PROJECT TITLE: BRUTE FORCE ATTACK INCIDENT RESPONSE ANALYSIS

PROJECT DESCRIPTION:

An incident response analysis of a simulated brute-force attack on an authentication system, demonstrating the identification, containment, and mitigation of security threats using the National Institute of Standards and Technology Cybersecurity Framework.

TOOLS AND FRAMEWORK USED:

1. Splunk
2. Google Docs
3. National Institute of Standards and Technology (NIST) Cybersecurity Framework.

SCENARIO:

Employees were unable to access internal systems due to a surge in failed login attempts. Investigation revealed a brute-force attack targeting user accounts from a suspicious external IP address. The incident was analyzed and handled using a structured response approach based on the National Institute of Standards and Technology Cybersecurity Framework.

INCIDENT RESPONSE ANALYSIS

SUMMARY

This morning, employees reported that they couldn't access internal systems. Suspicious login activity was detected, along with multiple failed login attempts.

IDENTIFY

The security team carried out an investigation and discovered multiple failed login attempts from the same IP address, indicating a brute-force attack on the authentication system. This attack was possible due to weak password policies, lack of monitoring, and no account lockout mechanism.

PROTECT

The incident response team responded by temporarily blocking suspicious IP addresses, disabling affected user accounts, enforcing password resets for all users, and restricting admin login access.

DETECT

To address the issue, the security team implemented the following:

- i. A SIEM tool (Splunk) to monitor login activity
- ii. Alerts for multiple failed login attempts
- iii. Log analysis to identify attack patterns

RESPOND

The security team was able to contain the attack. User accounts were secured, malicious IP addresses were blocked, and the systems were stabilized.

RECOVER

The security team restored systems to normal operation. Users regained access after password resets, monitoring systems were implemented, and security policies were updated.

NOTES:

This incident highlights the importance of account lockout policies, strong password policies, and continuous monitoring. It also shows that SIEM tools are essential for early detection of security threats.